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Single Mode Optical Fiber



LS Cable & System stands as a global leader in the telecommunications cable manufacturing sector and is recognized as one of the foremost market innovators. With years of experience serving one of the world's most technologically advanced markets, LS Cable & System possesses the expertise to provide efficient products tailored to your project needs. Our commitment to maintaining a high level of quality is widely acknowledged by our diverse clients, and our rigorous quality control processes consistently meet the stringent requirements of international standards.

Our comprehensive catalog is your guide to selecting the most efficient optical fibers.

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Long Haul, High-Capacity Networks (200~2,000km)

Fiber Name	ITU-T Standard	Description
PremierLink™ LE125	ITU-T G.654.E	Ultra Low Loss, Terrestrial Long-Haul and High-Capacity Fiber, Effective Area 125um
SimpleLink™ LWP	ITU-T G.652.D	Conventional Low Water Peak Fiber
PremierLink™ A1/A2	ITU-T G.652.D ITU-T G.657.A1/A2	Low Loss, MFD 9.2um, Bend Insensitive Fiber

Metro and Access Networks (~ 200km)

Fiber Name	ITU-T Standard	Description
SimpleLink™ LWP	ITU-T G.652.D	Conventional Low Water Peak Fiber
PremierLink™ A1/A2	ITU-T G.652.D ITU-T G.657.A1/A2	Low Loss, MFD 9.2um, Bend Insensitive Fiber
SimpleLink™ A1/A2	ITU-T G.652.D ITU-T G.657.A1/A2	Conventional Bend Insensitive Fiber

High Density Networks

Fiber Name	ITU-T Standard	Description
SimpleLink™ A1/A2 200	ITU-T G.652.D ITU-T G.657.A1/A2	200um Coated Conventional Bend Insensitive Fiber



Descriptions

PremierLink™ LE125 fiber, made from a pure silica core, has low attenuation and a large effective area, supporting terrestrial long-haul and high-capacity transmission links. It provides a suitable solution for long-reach 100 Gbps and beyond DWDM systems in terrestrial deployments by minimizing signal regeneration or amplification over long spans.

Standards

- ITU-T G.654.E
- IEC 60793-2-50 G.654.E

Applications

- Ultra long-haul network
- Repeaterless transmission systems for remote regions
- Optical transmission systems that require ultra low attenuation

Fiber Performance (Spec No. LSSS-OF0018)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1550nm	dB/km	≤ 0.170
	1625nm		≤ 0.190
Point Discontinuity	1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 30mm x 100 turn (at 1550/1625nm)	dB	≤ 0.1 / ≤ 0.1
Cable Cut Off Wavelength (λ _{cc})		nm	<1520
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	≤ 17 ~ 23
Mode Field Diameter	1550nm	μm	12.5 ± 0.5
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	≤ 0.050 ~ 0.070
PMD (Polarization Mode Dispersion)*	Fiber Individual	ps/√km	≤ 0.1
Group Index of Refraction	1550nm		1.462

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.8
Coating Diameter	μm	242 ± 5
Coating/Cladding Concentricity Error	μm	≤ 12
Fiber Curl (Radius of Curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	120
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _a)	-	≥ 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1550 nm and 1625 nm



Descriptions

SimpleLink™ LWP fiber is designed for full-spectrum optical transmission systems, it operates across the entire wavelength range from 1260nm to 1625nm. It is widely used in various communication networks, offering reliable performance and compatibility across various applications.

Standards

- ITU-T G.652.D
- IEC 60793-2-50 G.652.D

Applications

- Metro, Access network
- Long-haul network

Fiber Performance (Spec No. LSSS-OF0007)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.34
	1383nm		≤ 0.34
	1550nm		≤ 0.20
	1625nm		≤ 0.23
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 30mm x 100 turns (at 1550/1625nm)	dB	≤ 0.05 / ≤ 0.05
Cable Cut Off Wavelength (λ _{cc})		Nm	≤ 1260
Chromatic Dispersion Coefficients	1310nm	ps/(nm · km)	≤ 3.5
	1550nm		13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	9.1 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 0.9
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	242 ± 7
Coating/Cladding Concentricity Error	μm	< 12
Fiber Curl (Radius of Curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _d)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

SimpleLink™ A1 fiber is bend insensitive fiber in minimum 10mm of radius. It is fully compatible with G.652.D fiber and operates across the entire wavelength range from 1260nm to 1625nm. It provides resistance to bending loss and can be applied effectively in various applications installed in small or narrow space and can help to improve cable performance in high mechanical/ environmental requirement.

Standards

- ITU-T G.657.A1
- IEC 60793-2-50 G.657.A1

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0015)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.34
	1383nm		≤ 0.34
	1550nm		≤ 0.20
	1625nm		≤ 0.23
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.25 / ≤ 1.0
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.75 / ≤ 1.5
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	8.9 ± 0.4
Zero Dispersion Wavelength		Nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	242 ± 7
Coating/Cladding Concentricity Error	μm	< 12
Fiber Curl (Radius of Curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _d)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

SimpleLink™ A2 fiber is bend insensitive fiber in minimum 7.5mm of radius. It is fully compatible with G.652.D fiber and operates across the entire wavelength range from 1260nm to 1625nm. It provides high resistance to bending loss and can be applied effectively in various applications installed in small or narrow space and can help to improve cable performance in high mechanical/ environmental requirement.

Standards

- ITU-T G.657.A2
- IEC 60793-2-50 G.657.A2

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0014)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.34
	1383nm		≤ 0.34
	1550nm		≤ 0.20
	1625nm		≤ 0.23
Point Discontinuity	1310 nm and 1550 nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.03 / ≤ 0.1
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.1 / ≤ 0.2
	Radius 7.5mm x 1 turn (at 1550/1625nm)		≤ 0.5 / ≤ 1.0
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	8.6 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	242 ± 7
Coating/Cladding Concentricity Error	μm	< 12
Fiber Curl (Radius of Curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _a)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

PremierLink™ A1 fiber is bend insensitive fiber in minimum 10mm of radius and having low attenuation and 9.2μm mode field diameter. It provides additional margin and extends transmission reach, with high compatibility with G.652.D fiber. Furthermore, its high resistance to bending loss enables effective application in various installations within small or narrow spaces. This feature significantly enhances cable performance, particularly in environments with high mechanical or environmental demands.

Standards

- ITU-T G.657.A1 / G.652.D
- IEC 60793-2-50 G.657.A1

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0019)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.32
	1383nm		≤ 0.32
	1550nm		≤ 0.18
	1625nm		≤ 0.20
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.25 / ≤ 1.0
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.75 / ≤ 1.5
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	9.2 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	242 ± 7
Coating/Cladding Concentricity Error	μm	< 12
Fiber Curl (Radius of Curvature)	M	≥ 4

* A coating diameter of 200μm is also applicable

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _d)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

PremierLink™ A2 fiber is bend insensitive fiber in minimum 7.5mm of radius and having low attenuation and 9.2μm mode field diameter. It provides additional margin and extends transmission reach, with high compatibility with G.652.D fiber. Furthermore, its high resistance to bending loss enables effective application in various installations within small or narrow spaces. This feature significantly enhances cable performance, particularly in environments with high mechanical or environmental demands.

Standards

- ITU-T G.657.A2 / G.652.D
- IEC 60793-2-50 G.657.A2

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0020)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.32
	1383nm		≤ 0.32
	1550nm		≤ 0.18
	1625nm		≤ 0.20
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.03 / ≤ 0.1
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.1 / ≤ 0.2
	Radius 7.5mm x 1 turn (at 1550/1625nm)		≤ 0.5 / ≤ 1.0
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	9.2 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	242 ± 7
Coating/Cladding Concentricity Error	μm	< 12
Fiber Curl (Radius of Curvature)	M	≥ 4

* A coating diameter of 200μm is also applicable

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	1.3 ≤ F ≤ 8.9 (peak) 1.0 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _a)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05
			≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

SimpleLink™ A1 200 fiber is bend-insensitive fiber with a minimum bend radius of 10mm and a coating diameter of 200um. This reduced coating diameter increases fiber density in the cable and enables a compact cable size while maintaining the superior performance of SimpleLink™ A1 fiber. It is fully compatible with G.652.D fiber and operates across the entire wavelength range from 1260nm to 1625nm. Additionally, it meets and exceeds the standards of ITU-T G.657.A1 and G.652.D.

Standards

- ITU-T G.657.A1
- IEC 60793-2-50 G.657.A1

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0016)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.34
	1383nm		≤ 0.34
	1550nm		≤ 0.20
	1625nm		≤ 0.23
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.25 / ≤ 1.0
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.75 / ≤ 1.5
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	8.9 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope (S ₀)		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization Mode Dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding Non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	200 ± 7
Coating/Cladding Concentricity Error	μm	≤ 10
Fiber Curl (Radius of Curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	0.4 ≤ F ≤ 8.9 (peak) 0.4 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _d)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Descriptions

SimpleLink™ A2 200 fiber is bend insensitive fiber with a minimum bend radius of 7.5mm and a coating diameter of 200um. This reduced coating diameter increases fiber density in the cable and enables a compact cable size while maintaining the superior performance of SimpleLink™ A2 fiber. It is fully compatible with G.652.D fiber and operates across the entire wavelength range from 1260nm to 1625nm. Additionally, it meets and exceeds the standards of ITU-T G.657.A2 and G.652.D.

Standards

- ITU-T G.657.A2
- IEC 60793-2-50 G.657.A2

Applications

- Metro, Access network
- Long-haul network
- FTTx

Fiber Performance (Spec No. LSSS-OF0017)

Optical Performance			
Items	Conditions	Unit	Values
Attenuation	1310nm	dB/km	≤ 0.34
	1383nm		≤ 0.34
	1550nm		≤ 0.20
	1625nm		≤ 0.23
Point Discontinuity	1310nm and 1550nm	dB	≤ 0.05
Attenuation vs Bending	Radius 15mm x 10 turns (at 1550/1625nm)	dB	≤ 0.03 / ≤ 0.1
	Radius 10mm x 1 turn (at 1550/1625nm)		≤ 0.1 / ≤ 0.2
	Radius 7.5mm x 1 turn (at 1550/1625nm)		≤ 0.5 / ≤ 1.0
Cable Cut Off Wavelength (λ _{cc})		nm	≤ 1260
Chromatic Dispersion Coefficients	1550nm	ps/(nm · km)	13.3 ≤ D ≤ 18.6
	1625nm		17.2 ≤ D ≤ 23.7
Mode Field Diameter	1310nm	μm	8.6 ± 0.4
Zero Dispersion Wavelength		nm	1300 ~ 1324
Zero Dispersion Slope		ps/(nm ² · km)	0.073 ≤ S ₀ ≤ 0.092
PMD (Polarization mode dispersion)*	Link design value	ps/√km	≤ 0.1
Group Index of Refraction	1310 / 1550 / 1625nm	-	1.4667 / 1.4683 / 1.4683

*PMD value may change when fiber is cabled. Cabled PMD is recommended to be verified by cable manufacturer.

Geometrical Performance		
Items	Unit	Values
Cladding Diameter	μm	125 ± 0.7
Cladding non-Circularity	%	≤ 1.0
Core/Cladding Concentricity Error	μm	≤ 0.5
Coating Diameter	μm	200 ± 7
Coating/Cladding Concentricity Error	μm	≤ 10
Fiber Curl (Radius of curvature)	M	≥ 4

Mechanical Performance		
Items	Unit	Values
Proof Stress Level	Kpsi	≥ 100
Coating Strip Force	N	0.4 ≤ F ≤ 8.9 (peak) 0.4 ≤ F ≤ 5.0 (average)
Dynamic Fatigue Resistance Aged and Unaged (N _d)	-	> 20

Environmental Performance			
Items	Conditions	Unit	Values
Temperature Cycling	-60°C to +85°C	dB/km	≤ 0.05
Temperature-Humidity Cycling	-10°C to +85°C and 98% RH		≤ 0.05
Water Immersion	23°C, 30 days		≤ 0.05
Dry Heat	85°C, 30 days		≤ 0.05
Damp Heat	85°C, 85% RH, 30 days		≤ 0.05

* Induced attenuation change at 1310 nm, 1550 nm and 1625 nm



Single mode optical fiber

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